OSSE Observation-Value Report

Nature run, synthetic sampling, Green's-function sensitivity,
and channel ranking for the FORTE Expedition field-campaign plan

Four parts, end to end:

1. Nature run + 5 synthetic observation channels matching the FORTE campaign's real instrument plan as closely as a 1-D model allows.
2. Green's-function sensitivity: how the model state responds to a perturbation in each of 11 boundary/forcing inputs.
3. The observation operator connecting (1) and (2) -- which model variables and controls each channel actually informs.
4. The ranking itself: a Fisher-information/degrees-of-freedom-for-signal score per channel, answering the actual research question -- which observation TYPE most reduces uncertainty in the model's inputs.

Built by tools/run_osse_nature.sh, build_osse_observations.py, build_green_perturbations.py, build_green_functions.py, and rank_observation_value.py -- each documented in full in its own module docstring.

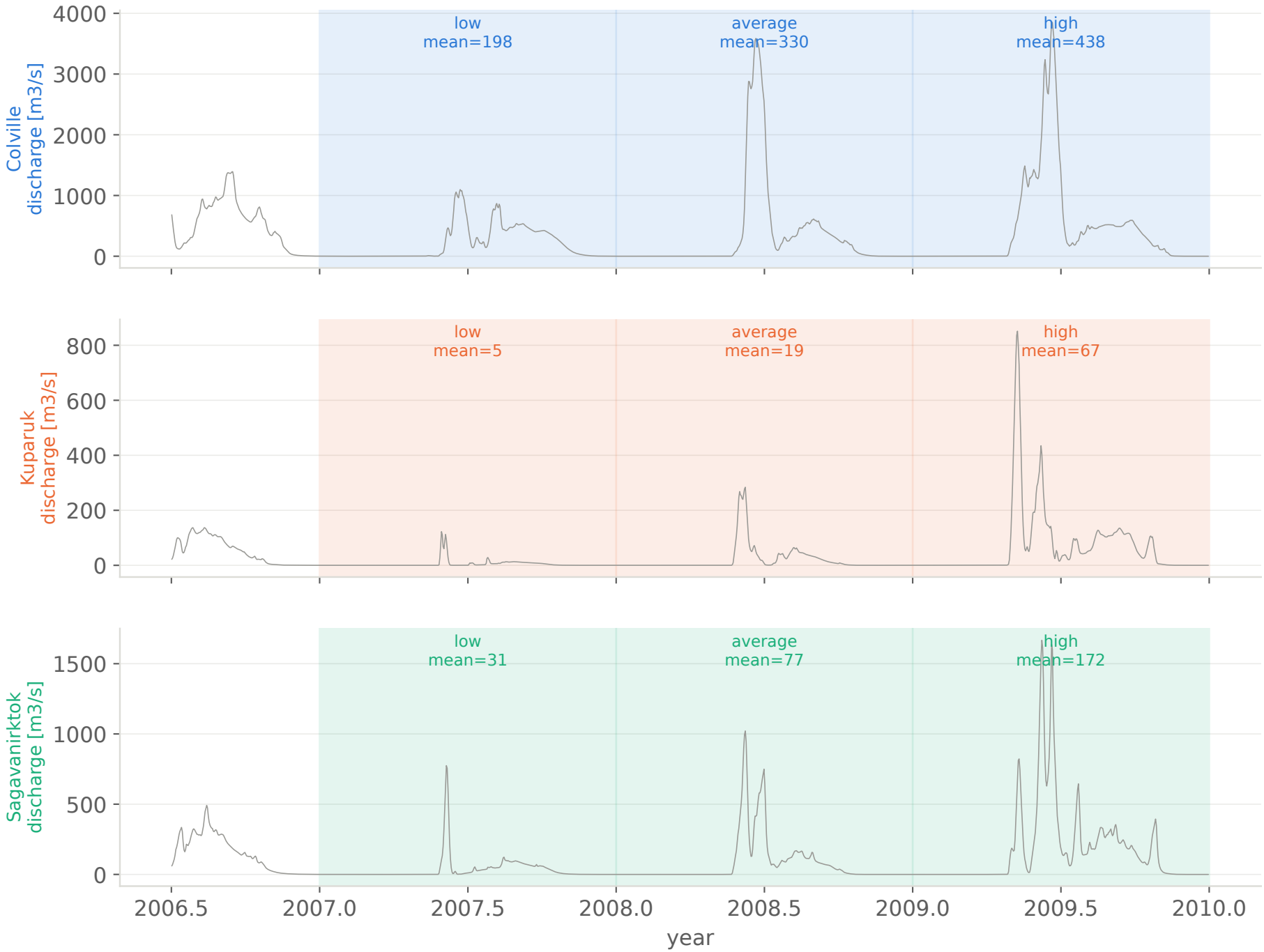
FORTE campaign phases -> OSSE day-of-year windows

Phase	Real dates (2027)	Day-of-year	Instruments	Notes
Phase 0	Late April (TBD)	--	Under-ice	NOT SYNTHESIZED -- source doc marks infeasible (negligible under-ice flow)
Phase 1a	Jun 1-5	152-156	Grab sampling	Helicopter, pre-breakup; DOC/nutrients/DIC/ALK/Chl-a
Phase 1b	Jun 6-20	157-171	Grab sampling + ice drone	Helicopter+boats; ice-elevation drone work NOT synthesized (different product)
Phase 2	Jul 1-12	182-193	AVIRIS-3 + moorings deployed	Boats+R/V Ukpik; moorings go in, stay through Phase 3
Phase 3	Jul 20-Aug 7	201-219	AVIRIS-3+drone, offshore moorings	Boats+R/V Ukpik+OceanXplorer

Real campaign is 2027 (Phase 0 "maybe 2028"); NS-RAD's forcing stops at 2023 and the model has no restart/checkpoint capability, so these windows are mapped onto three ANALOG years below, not literally 2027.

Nature-run analog years: 2007 (low) / 2008 (average) / 2009 (high)

Chosen as the cleanest consecutive low/average/high flow-regime triplet in the 2005-2023 record for all three rivers simultaneously -- the real 2027/2028 campaig

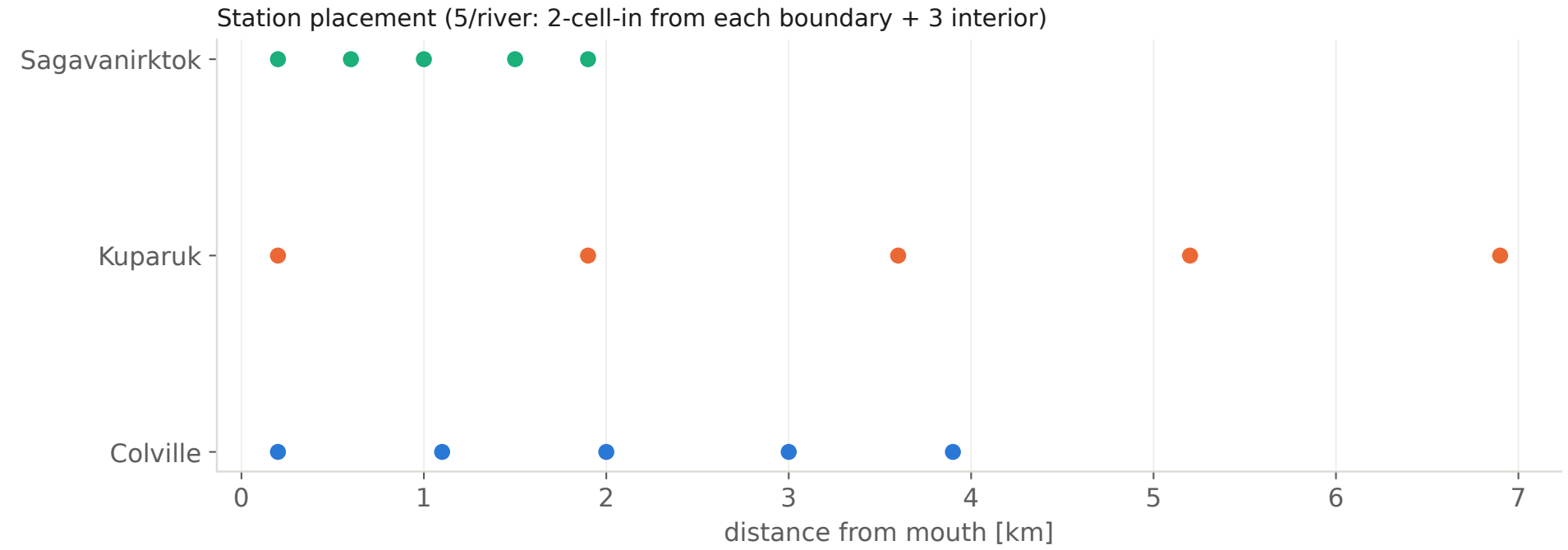


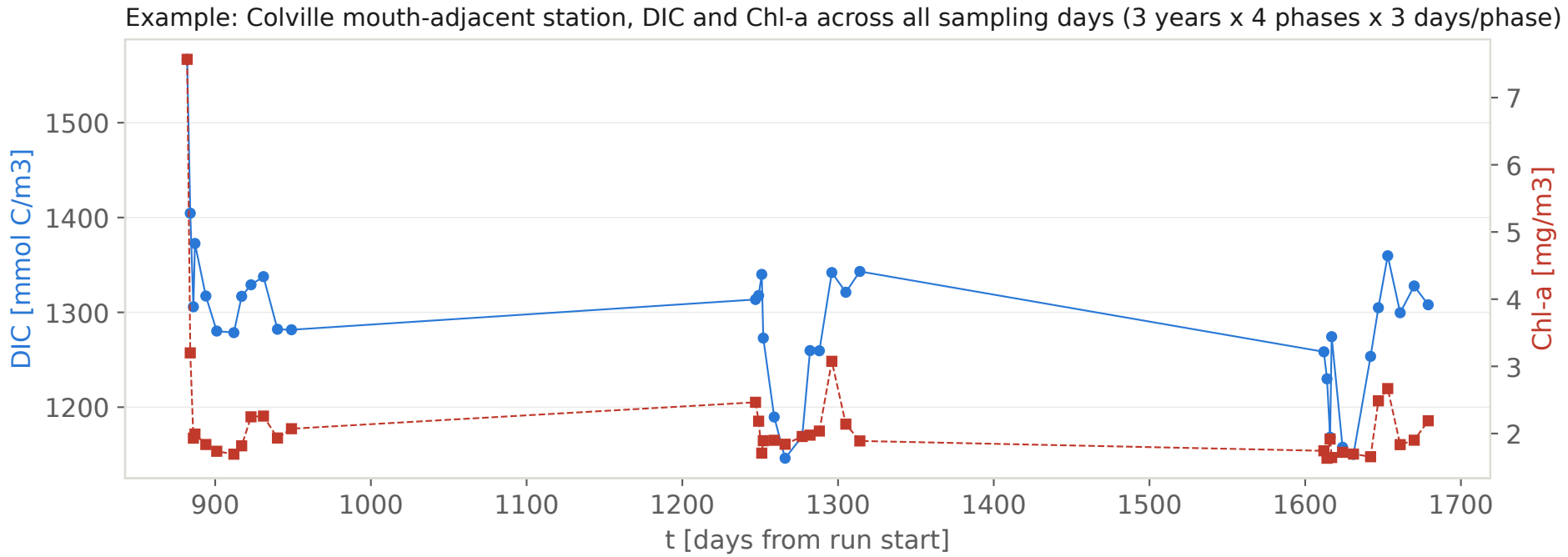
Five observation channels: design summary

Channel	Spatial pattern	Temporal pattern	Variables	Error
Grab sampling	5 stations/river (2-cell-in from each boundary + 3 interior)	4 phases x 3 days/phase (first/mid/last)	DOC,CDOM(proxy),NH4, NO3,PO4,dSi,DIC,ALK, Chl-a(DIA proxy)	2-15% (species-dependent)
Moorings	1 interior (inshore) + boundary forcing (offshore)	Continuous, Jul1-Sep30 (deployed/recovered)	S, T, O2, current speed (inshore); S,T,O2 (offshore)	T+-0.2C, S+-0.1psu, O2+-5%
Drifters	Lagrangian (advected by model velocity field)	30/year, released in 4 batches (10/10/5/5) by campaign phase	S, T along trajectory + arrival time at mouth	T+-0.3C, S+-0.2psu
Airborne (AVIRIS-3+drone)	Every grid cell (full longitudinal snapshot)	Every 8h, Phase 2+3 only (not Phase 1)	CDOM proxy, Chl-a proxy, SPM (optical only)	15-20%
Satellite (SWOT+PACE, notional)	SWOT: width>=100m cells; PACE: outer 1-2 cells only	SWOT ~21-day repeat; PACE ~daily, 50% cloud dropout	SWOT: depth anomaly; PACE: CDOM/Chl-a proxy	SWOT 3-10cm (width-dep.); PACE 25%

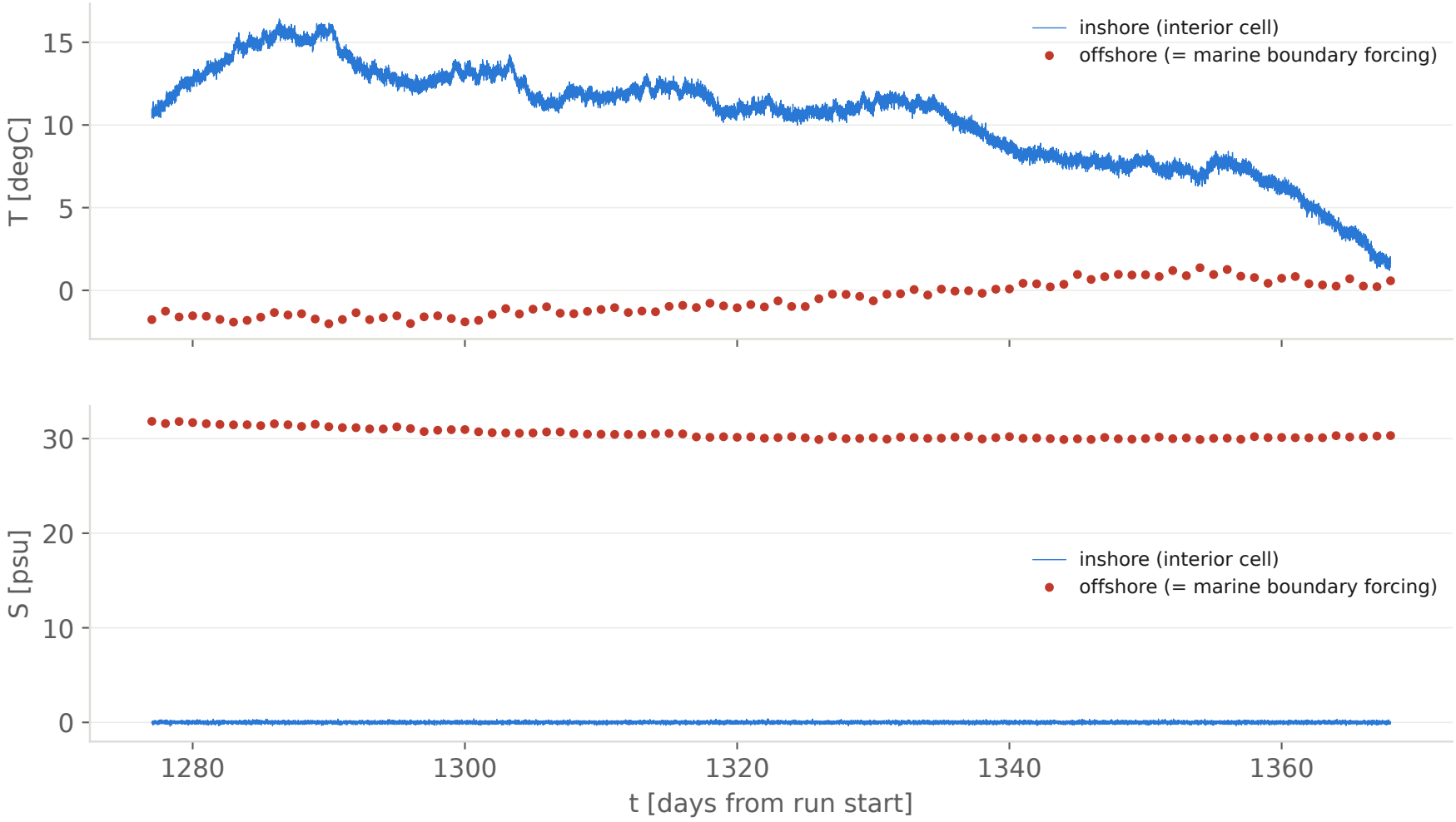
NOT synthesized (documented gaps, not oversights): POC/PON (no defensible NS-RAD state-variable proxy), flow cytometry/productivity incubations (process/community measurements, no model analog), Phase 1b ice-elevation drone work (a different product). Satellite channel gated by PHYSICAL limits only (ice, cloud, channel width), not real launch dates -- see module docstring for why.

Grab sampling -- station placement and example time series



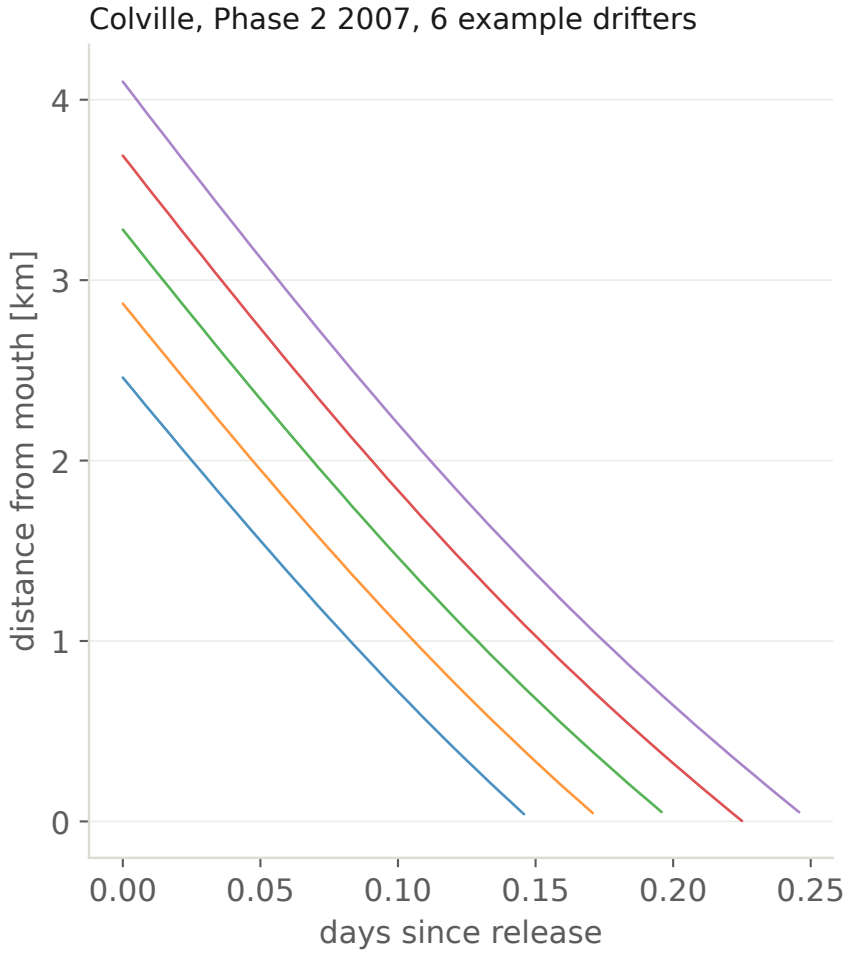


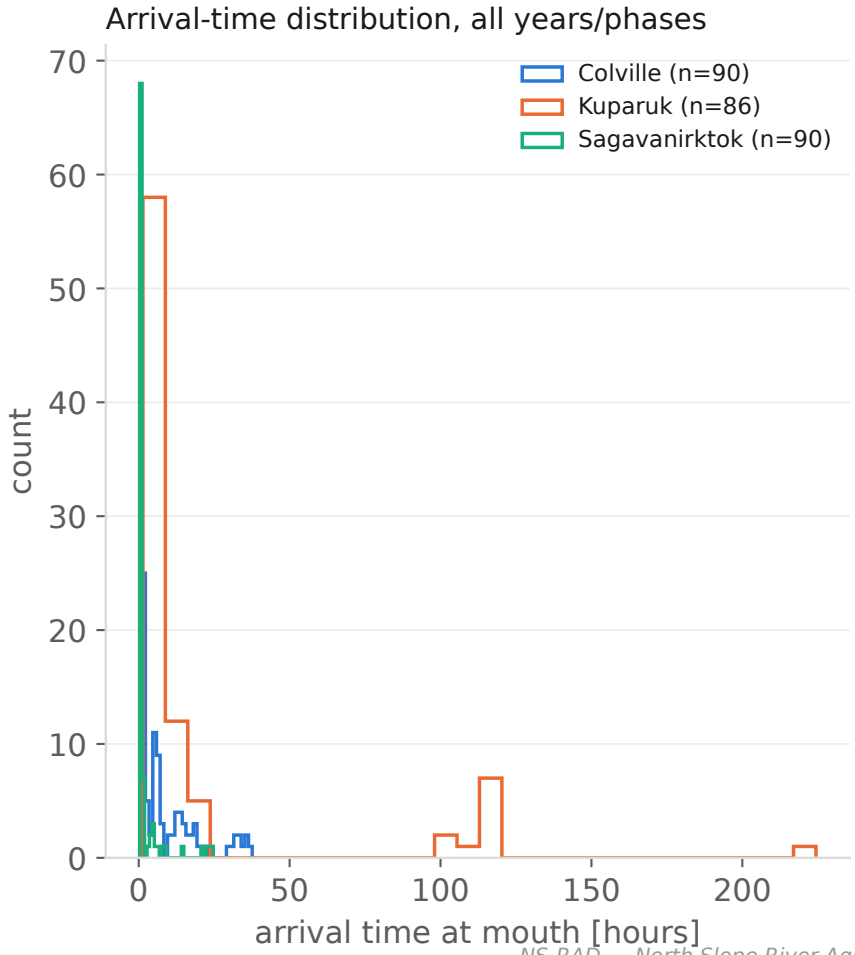
Moorings -- Colville, inshore vs. offshore, 2008 (average year)



Offshore samples the marine BOUNDARY FORCING file directly (daily native cadence) -- it validates the assumed boundary input, not interior model skill, a conceptually different product from the inshore mooring despite the shared instrument class.

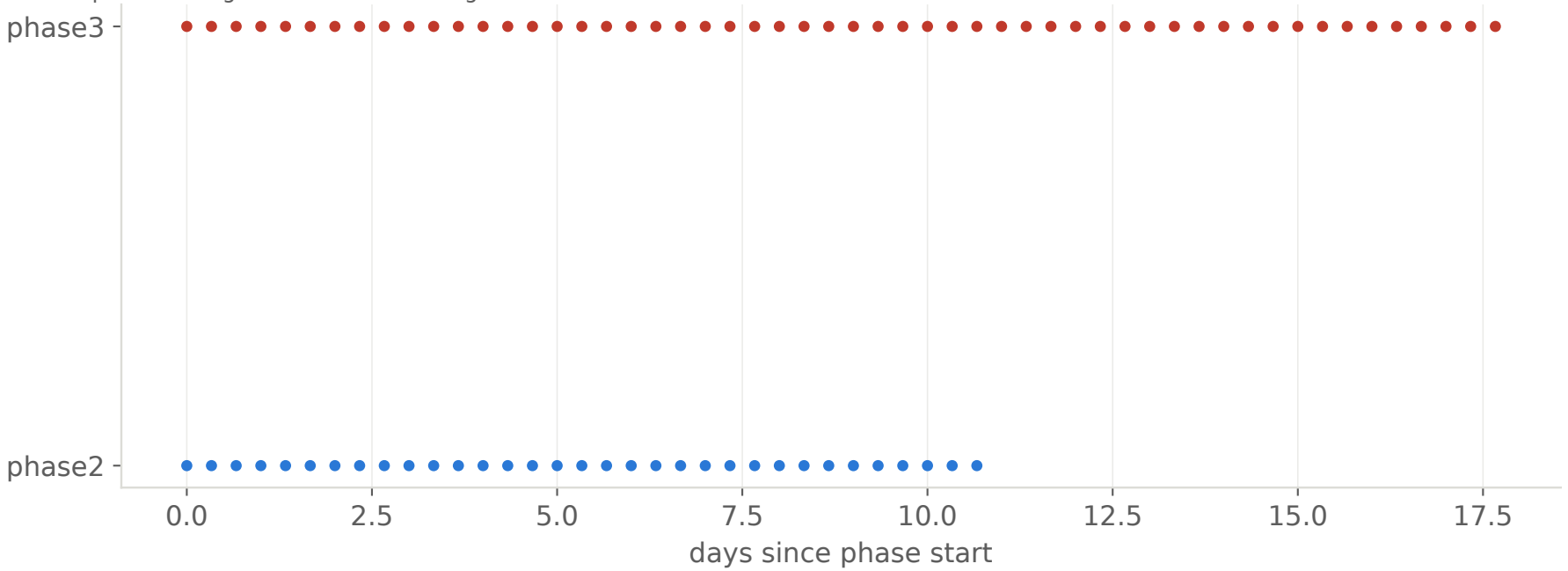
microSWIFT drifters -- example trajectories and arrival-time distribution





Airborne (AVIRIS-3+drone) -- Colville 2008 flight times

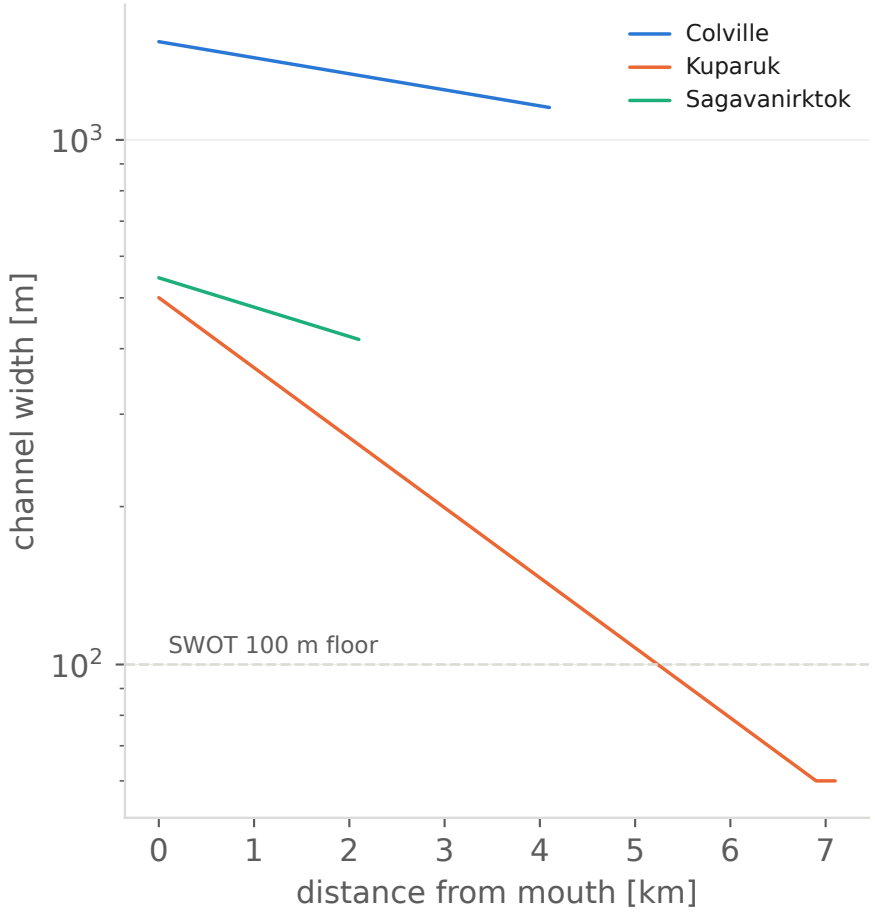
Every 8h within each phase window -- 8h does not divide evenly into any dominant tidal period (M2/S2/O1/K1), so sampled tidal phase keeps advancing rather than re-locking.



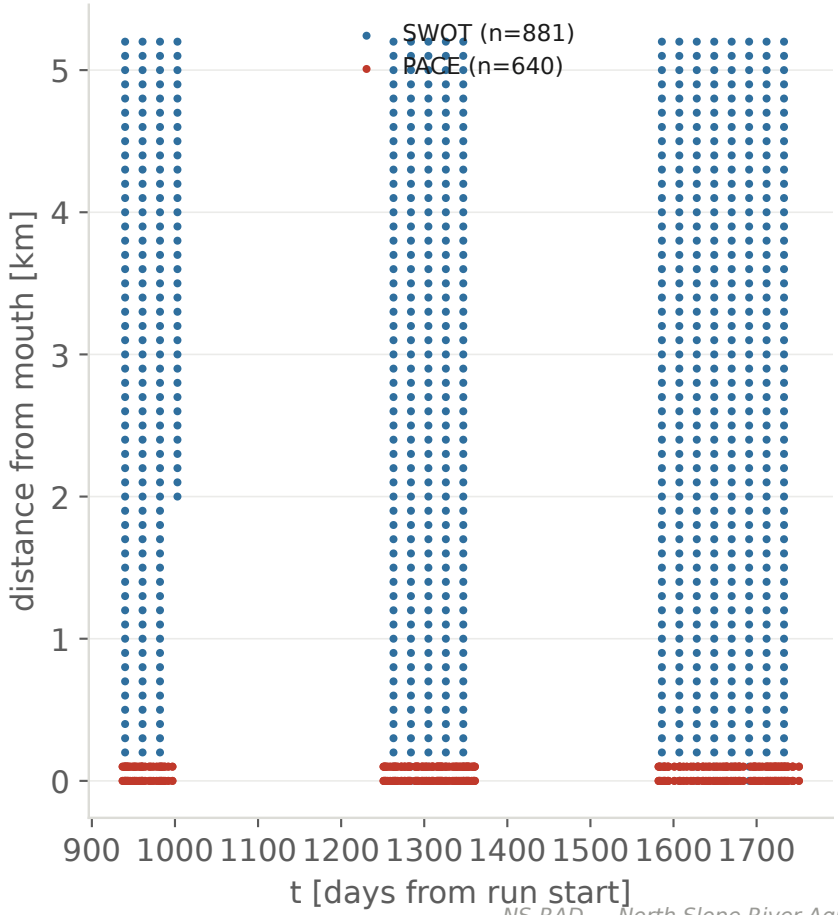
Phase 2 (12 days) and Phase 3 (19 days) both approach or exceed the ~14.8-day M2/S2 spring-neap beat period, so spring-neap diversity falls out of the real campaign windows without special-case scheduling.

Satellite (notional): SWOT width-resolvability and coverage

All three rivers flare from a narrow upstream reach to a wide mouth -- SWOT's blind spot is upstream-only, not whole-river



Kuparuk: coverage gaps = ice season (+ PACE cloud dropout)



Synthetic observation row counts, by river x channel

River	Grab	Moorings	Drifters	Airborne	Satellite
Colville	1,620	262,920	18,540	32,886	1,344
Kuparuk	1,620	262,920	54,776	56,376	1,521
Sagavanirktok	1,620	262,920	4,292	17,226	1,234

Green's-function sensitivity: methodology

Control	What it perturbs
discharge	River discharge forcing
clb_T	Marine boundary: sea temperature
clb_S	Marine boundary: S
clb_DIC	Marine boundary: DIC
clb_ALK	Marine boundary: ALK
clb_NO3	Marine boundary: NO3
clb_NH4	Marine boundary: NH4
clb_PO4	Marine boundary: PO4
clb_dSi	Marine boundary: dSi
clb_O2	Marine boundary: O2
clb_TOC	Marine boundary: TOC

11 CONTROLS per river (table, left): discharge and the marine boundary for every genuinely time-varying solute in the full-forcing variant. Upstream (cub) boundaries are NOT perturbed -- every solute except TOC holds its upstream boundary constant even in the full-forcing variant, so there is nothing time-varying there to be uncertain about.

PERTURBATION: a single one-sided +10% relative step applied to the WHOLE forcing series (not a delta-impulse) -- this answers "if this boundary were systematically biased 10% high, how would the model respond," the right question for ranking which OBSERVATIONS best catch a persistent forcing misspecification.

NS-RAD has no adjoint (a ported legacy Fortran model, not auto-differentiable), so sensitivity is FINITE-DIFFERENCE: one baseline and one perturbed run per control, differenced and divided by the 10% step:

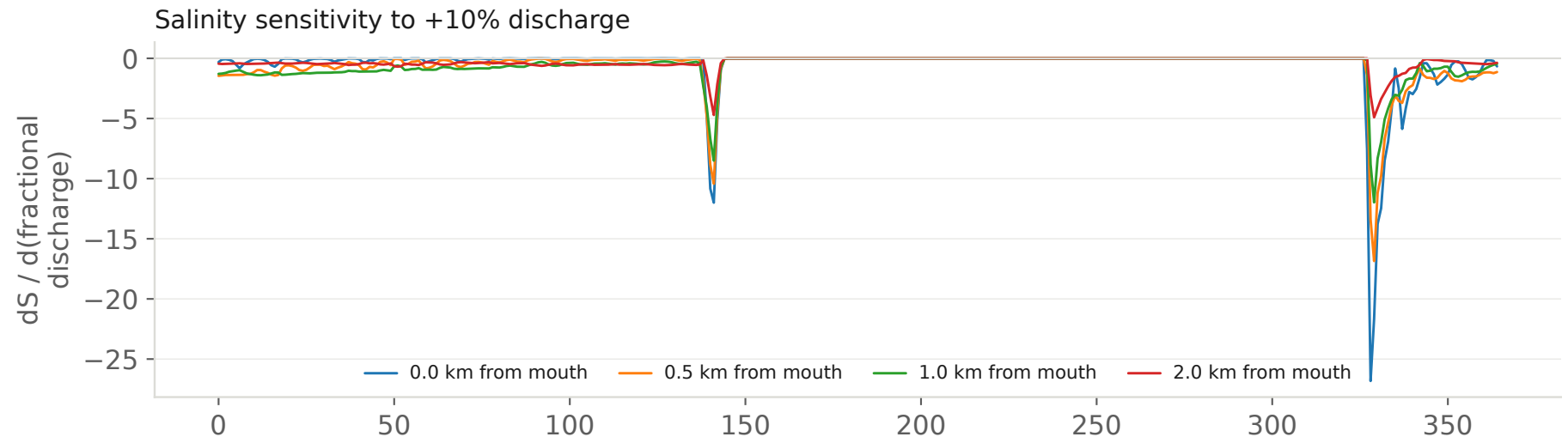
$$G_c(doy, cell) = [X_{perturbed}(doy, cell) - X_{baseline}(doy, cell)] / 0.10$$

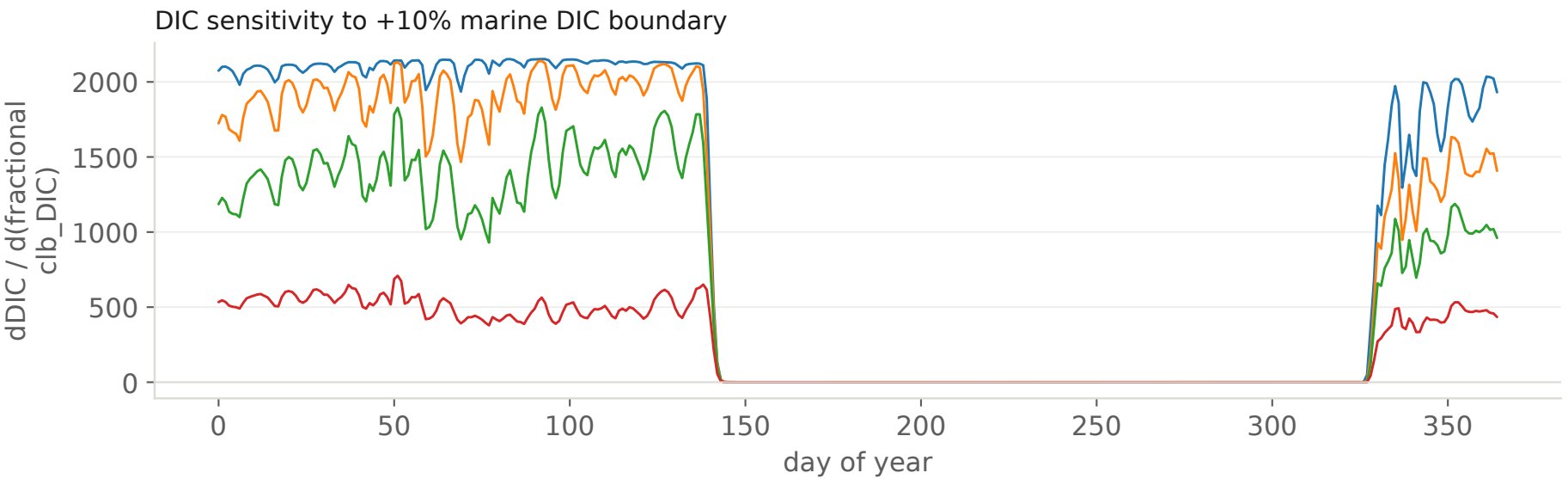
COST: each perturbation run is MAXT_DAYS=730 (365 warmup + 365 post-warmup reference year, 2006) at DAILY cadence -- cheap (~minutes), since Green's functions only need the seasonal SHAPE of the model's sensitivity, not tidal-scale dynamics. 2006 is used because NS-RAD has no restart/checkpoint capability -- it is the first full year reachable at all. 3 rivers x 11 controls = 33 perturbation runs, plus 3 baselines = 36 total.

KEY CAVEAT: sensitivity is assumed STATIONARY IN DAY-OF-YEAR across different calendar years -- i.e. "how a +10% DIC bias in May propagates" is assumed similar in 2006, 2007, 2008, and 2009, even though the actual flow regime differs across those three analog study years (that is the whole reason they were chosen as low/avg/high). A fully rigorous version would compute a SEPARATE reference run per analog year -- not done here, due to the same no-restart cost problem that made reaching 2007/2008/2009 themselves expensive in the first place. This is the single most important caveat for interpreting the ranking results in Part 4.

Example sensitivity fields -- Colville, reference year 2006

Finite-difference sensitivity dX/d(fractional control), by day-of-year and distance from mouth -- both panels reveal the SAME real, physically coherent pattern: marine/discharge influence collapses when the estuary is fully fresh.





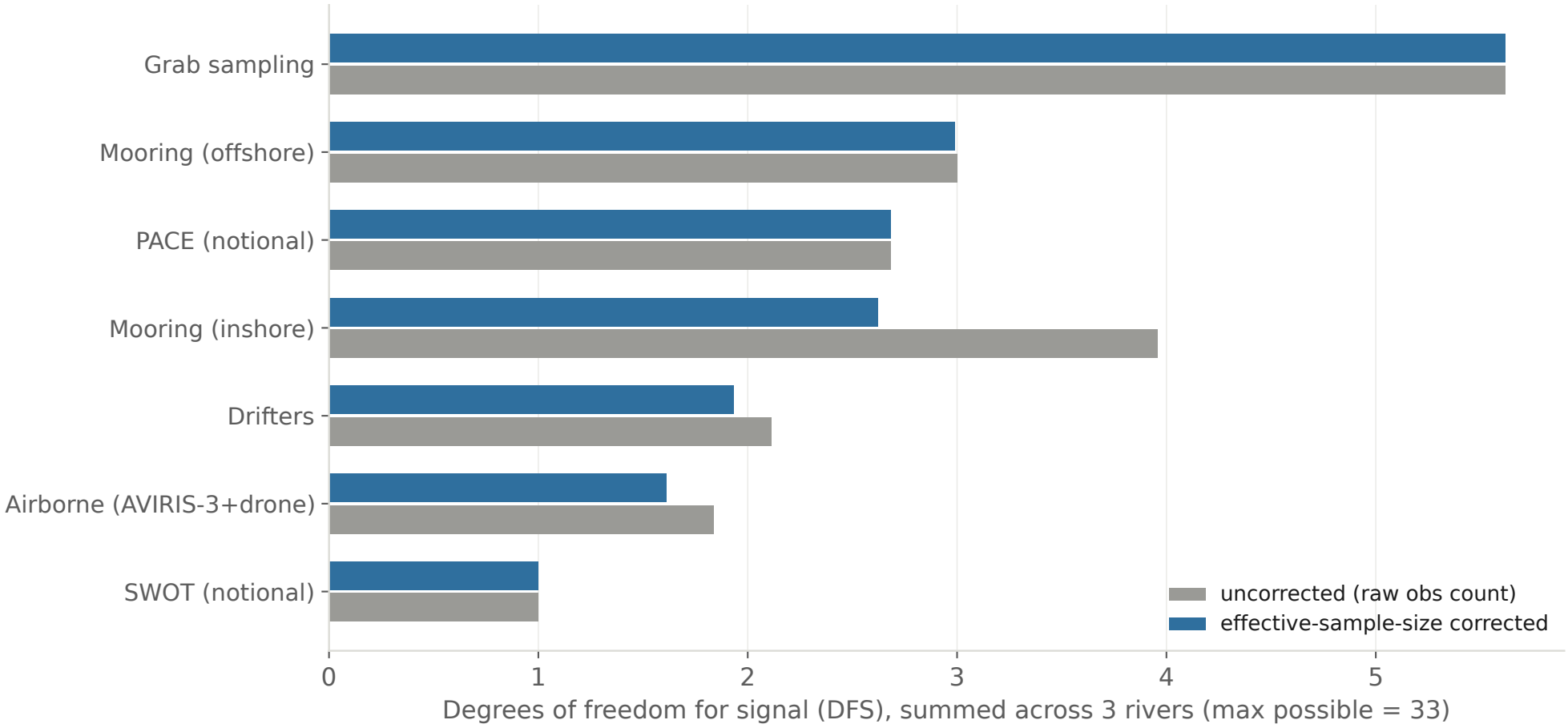
Both sensitivities are non-zero only while the estuary is salt-intruded (winter/low-flow); during the spring freshet the domain flushes completely fresh and baseline==perturbed exactly (verified directly against the raw netCDF, not a plotting artifact) -- the marine boundary genuinely cannot influence a cell with zero residual marine water in it.

Observation-to-control mapping: what each channel can inform

Channel	Model variables observed	Controls it can inform	Note
Grab sampling	ALK, DIA, DIC, NH4, NO3, PO4, TOC, dSi	clb_ALK, clb_DIC, clb_NH4, clb_NO3, clb_PO4, clb_TOC, clb_dSi, discharge	
Mooring (inshore)	O2, S, T	clb_O2, clb_S, clb_T, discharge	
Mooring (offshore)	S, T, O2	clb_S, clb_T, clb_O2	reads the marine boundary FORCING file directly (not a propagated model field) -- validates the assumed input itself
Drifters	S, T	clb_S, clb_T, discharge	
Airborne (AVIRIS-3+drone)	DIA, SPM, TOC	clb_TOC, discharge	
SWOT (notional)	depth	discharge	
PACE (notional)	DIA, TOC	clb_TOC, discharge	

"Controls it can inform" = model variables that channel samples AND that have a corresponding Green's-function control (CDOM/Chl-a proxies map to TOC/DIA, per tools/rank_observation_value.py's VAR_MAP). A channel reaching a control does not guarantee it meaningfully CONSTRAINS it -- see Part 4 for the actual information content.

Observation-value ranking: which channel most reduces boundary/forcing uncertainty



Mooring (inshore) drops from #2 to #4 once same-day/sub-daily repeat samples (which share an IDENTICAL daily-resolution Green's-function row) are no longer double-counted as independent evidence -- the single biggest qualitative change this correction makes.

Effective-sample-size correction, by channel

Channel	Raw observations (summed, 3 rivers)	Effective observations (summed, 3 rivers)	Overcounting factor
Grab sampling	4,860	4,860	1.0x
Mooring (inshore)	589,707	2,484	237.4x
Mooring (offshore)	2,484	108	23.0x
Drifters	77,342	3,794	20.4x
Airborne (AVIRIS-3+drone)	106,488	37,944	2.8x
SWOT (notional)	1,971	1,971	1.0x
PACE (notional)	2,128	2,128	1.0x

Two caps (tools/rank_observation_value.py): (A) MECHANICAL, every channel -- the Green's-function arrays are daily-resolution only, so same-day repeat samples of the same (variable, cell) share an identical sensitivity row; (B) PHYSICAL, mooring (offshore) only -- it reads the marine boundary forcing directly, whose real update rate is ~monthly (ECCO-Darwin v5), so daily-interpolated samples within the same ~30-day window add no new information.

Which controls each channel actually constrains (corrected, summed across 3 rivers)

Channel	Top 3 constrained controls (variance reduction, prior = 20% rel. uncertainty)
Grab sampling	discharge (0.0400), clb_ALK (0.0390), clb_NO3 (0.0382)
Mooring (inshore)	discharge (0.0399), clb_S (0.0385), clb_O2 (0.0207)
Mooring (offshore)	clb_S (0.0400), clb_O2 (0.0399), clb_T (0.0396)
Drifters	discharge (0.0399), clb_S (0.0370), clb_T (0.0004)
Airborne (AVIRIS-3+drone)	discharge (0.0400), clb_TOC (0.0213), clb_PO4 (0.0022)
SWOT (notional)	discharge (0.0400)
PACE (notional)	discharge (0.0400), clb_PO4 (0.0343), clb_dSi (0.0250)

Summary: observation-value ranking

1. Grab sampling -- DFS = 5.62
2. Mooring (offshore) -- DFS = 2.99
3. PACE (notional) -- DFS = 2.68
4. Mooring (inshore) -- DFS = 2.62
5. Drifters -- DFS = 1.93
6. Airborne (AVIRIS-3+drone) -- DFS = 1.61
7. SWOT (notional) -- DFS = 1.00

Key findings:

- Grab sampling is the clear top channel -- sparse but spans many species, stations, and phases, reaching essentially every control.

- Mooring (offshore) and PACE are surprisingly competitive: both are INFORMATION-EFFICIENT -- they reach full value from relatively few truly-independent samples, unlike continuous in-situ monitoring.

- Mooring (inshore) ranks lower than its raw observation count would suggest once duplicate same-day samples are no longer double-counted (Part 4's effective-sample-size correction).

- SWOT (notional) ranks lowest -- its depth-anomaly signal reaches only the discharge control, not any biogeochemical boundary.

All results carry the Part 2 stationarity caveat (sensitivity built from ONE reference year, 2006, reused for 2007-2009) and the single, equal 20% relative-uncertainty prior on every control -- see those sections before treating small DFS differences as decisive.